

Q1. List Mutation

Write a function `remove_last(lst)` that removes the last element of a list.

Call this function with a list and check whether the original list changes outside the function.

Q2. String Reassignment

Write a function `change_string(s)` that tries to replace the first character of the string with "X".

Call this function and check whether the original string changes outside the function.

Q3. Dictionary Mutation vs Rebinding

(a) Write a function `add_entry(d)` that adds a new key-value pair into a dictionary.

(b) Write another function `reassign_dict(d)` that reassigns the dictionary variable to a new dictionary.

Test both functions on the same dictionary and observe the results.

Q4. Write a function to find the maximum of two numbers.

- Input: two numbers
- Output: the greater number

Q5. Write a function to calculate the factorial of a number.

- Input: an integer n
- Output: factorial value of n

Q6. Write a function to convert temperature from Celsius to Fahrenheit.

- Formula: $F = (C \times 9/5) + 32$
- Input: temperature in Celsius
- Output: temperature in Fahrenheit